

Exp : 4 Backup Automation using Azure Blob Storage

Aim

To design and implement an automated backup solution using Azure Blob Storage and scheduling tools.

Azure Blob Storage is a cloud-based object storage service used to store large amounts of unstructured data such as files, backups, logs, and media. It provides high availability, scalability, and secure storage for backup and disaster recovery solutions.

Procedure

Step 1: Create Azure Storage Account

1. Login to Azure Portal: <https://portal.azure.com>
2. Click **Create a resource**
3. Search **Storage Account** → Click **Create** Fill the following details:
 - Subscription: Azure Student Subscription
 - Resource Group: Create new (backup-rg)
 - Storage Account Name: mystoragebackup123
 - Region: South India
 - Performance: Standard
 - Redundancy: Locally Redundant Storage (LRS)
4. Click **Review + Create** → **Create**

Step 2: Create Blob Container

1. Open Storage Account
2. Go to **Data Storage** → **Containers**
3. Click + **Container**

Fill:

- Name: backups
 - Public Access: Private
4. Click **Create**

Step 3: Azure CLI Login

```
az login
# Opens browser for authentication
```

```
az account show
# Verify account details
```

Step 4: Upload Files

```
az storage blob upload-batch \  
  --account-name mystoragebackup123 \  
  --destination backups \  
  --source C:\backup-data  
# Uploads all local files to Azure
```

Step 5: Create Backup Script

```
$source = "C:\backup-data"  
$container = "backups"  
  
az storage blob upload-batch \  
  --account-name mystoragebackup123 \  
  --destination $container \  
  --source $source  
# Automates backup process
```

Step 6: Add Timestamp Backup

```
$date = Get-Date -Format "yyyy-MM-dd"  
$destination = "backups/$date"  
  
az storage blob upload-batch \  
  --account-name mystoragebackup123 \  
  --destination $destination \  
  --source C:\backup-data  
# Stores backup by date
```

Step 7: Scheduling using Windows Task Scheduler (Detailed)

Steps:

1. Press **Windows + R**
2. Type:
taskschd.msc
3. Press Enter
4. Click **Create Basic Task**
5. Enter:
 - Name: AzureBackupTask
 - Description: Daily Azure Backup
6. Click **Next**

Set Trigger

- Choose: **Daily**
- Set time (e.g., 2:00 AM)
- Click Next

Set Action

- Choose: **Start a Program**

Configure Program

powershell.exe

Add Arguments:

-ExecutionPolicy Bypass -File "C:\scripts\backup.ps1"

Start In:

C:\scripts

Finish

- Click Next → Finish

OUTPTU:

Step 1: Storage Account Created Successfully

Home > Storage accounts > mystoragebackup123

Overview

Resource group (selected): backup-rg

Location: South India

Primary/Secondary Location: Primary: South India

Subscription (selected): Azure for Students

Subscription ID: 00000000-0000-0000-0000-000000000000

Disk state: Available

Tags (info)

Step 2: Container Created Successfully

Home > mystoragebackup123 > Containers > backups

Overview

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: backups

Search blobs by prefix (case-sensitive)

Add filter

Name Modified Access tier Archive status

No results

Step 3: Azure CLI Login Output

```
PS C:\Users\student> az login
A browser has been opened at https://login.microsoftonline.com/
o sign in.
Retrieving tenants and subscriptions for the selection...
[
  {
    "cloudName": "AzureCloud",
    "homeTenantId": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
    "id": "11111111-2222-2233-4444-555555555555",
    "isDefault": true,
    "name": "Azure for Students",
    "state": "Enabled",
    "tenantId": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
    "user": {
      "name": "student@example.com",
      "type": "user"
    }
  }
]
PS C:\Users\student>
```

Step 4: Upload Files to Blob (CLI Output)

```
PS C:\Users\student> az storage blob upload-batch \
--account-name mystoragebackup123 \
--destination backups \
--source C:\backup-data
Finished[#####] 100.0000%
Successfully uploaded 3 files.
PS C:\Users\student>
```

Step 5: Run Backup Script Manually

```
PS C:\scripts> .\backup.ps1
Finished[#####] 100.0000%
Successfully uploaded 3 files.
PS C:\scripts>
```

Step 6: Timestamp Backup (CLI Output)

```
PS C:\Users\student> $date = Get-Date -Format "yyyy-MM-dd"
PS C:\Users\student> $destination = "backups/$date"
PS C:\Users\student> az storage blob upload-batch \
--account-name mystoragebackup123 \
--destination $destination \
--source C:\backup-data
Finished[#####] 100.0000%
Successfully uploaded 3 files.
PS C:\Users\student>
```

Step 7: Task Scheduler (Backup Scheduled)

Create Basic Task Wizard

Name: AzureBackupTask

Trigger: Daily

Action: Start a Program

Trigger: Daily At 02:00 every day

Action: Start a program: powershell.exe -ExecutionPolicy Bypass -File "C:\scripts\backup.ps1"

Verify: Files in Azure Container

Name	Modified	Access tier	Archive status	Blob type	Size
2025-05-11\	5/11/2025, 2:00:16 AM	Hot		Block blob	3 files
2025-05-12\	5/12/2025, 2:00:15 AM	Hot		Block blob	3 files
2025-05-13\	5/13/2025, 2:00:14 AM	Hot		Block blob	3 files

Result

Successfully implemented automated backup using Azure Blob Storage with scheduling.